



Initial Project Proposal

(Counselling Session Booking System – Group 16)

CSE3023 Web-Based Application Development

Bachelor of Computer Science (Software Engineering)

Matric ID	Name
S74991	WAI PEI YUEN
S61748	TENGKU AQIL BIN TENGKU AMDAN
S74232	AMERRUL RAFIQUE BIN MOHAMED SULAIMAN
S75851	MUHAMMAD ADAM DANISH BIN MOHAMAD FAIZ

1.0 Introduction to the Target Project

1.1 Project Title

Counselling Session Booking System

1.2 Project Overview

The Counselling Session System is a dynamic web-based application. It developed by using Model View-Controller (MVC) with java web technologies. It is designed to provide a centralized platform to help students book counselling sessions online easily and efficiently.

The system allows users to register and log in, view counsellor scheduels, book counselling sessions, manage booking and provide feedback after sessions. For the counsellors, they can manage their schedules and record counselling sessions with notes while administrators can manage users, counsellors and monitor system activities. For each main module, we will implement CRUD operations.

1.3 Problem statement

Currently, counseling session bookings are often handled manually or through informal methods like messaging, which can lead to scheduling conflicts and double booking, lack of proper records and inconvenience for both students and counselors.

Therefore, a centralized system is needed to manage appointments efficiently, reduce errors, improve accessibility and improve proper records.

2.0 Group Members, Leader and Module Task Distribution

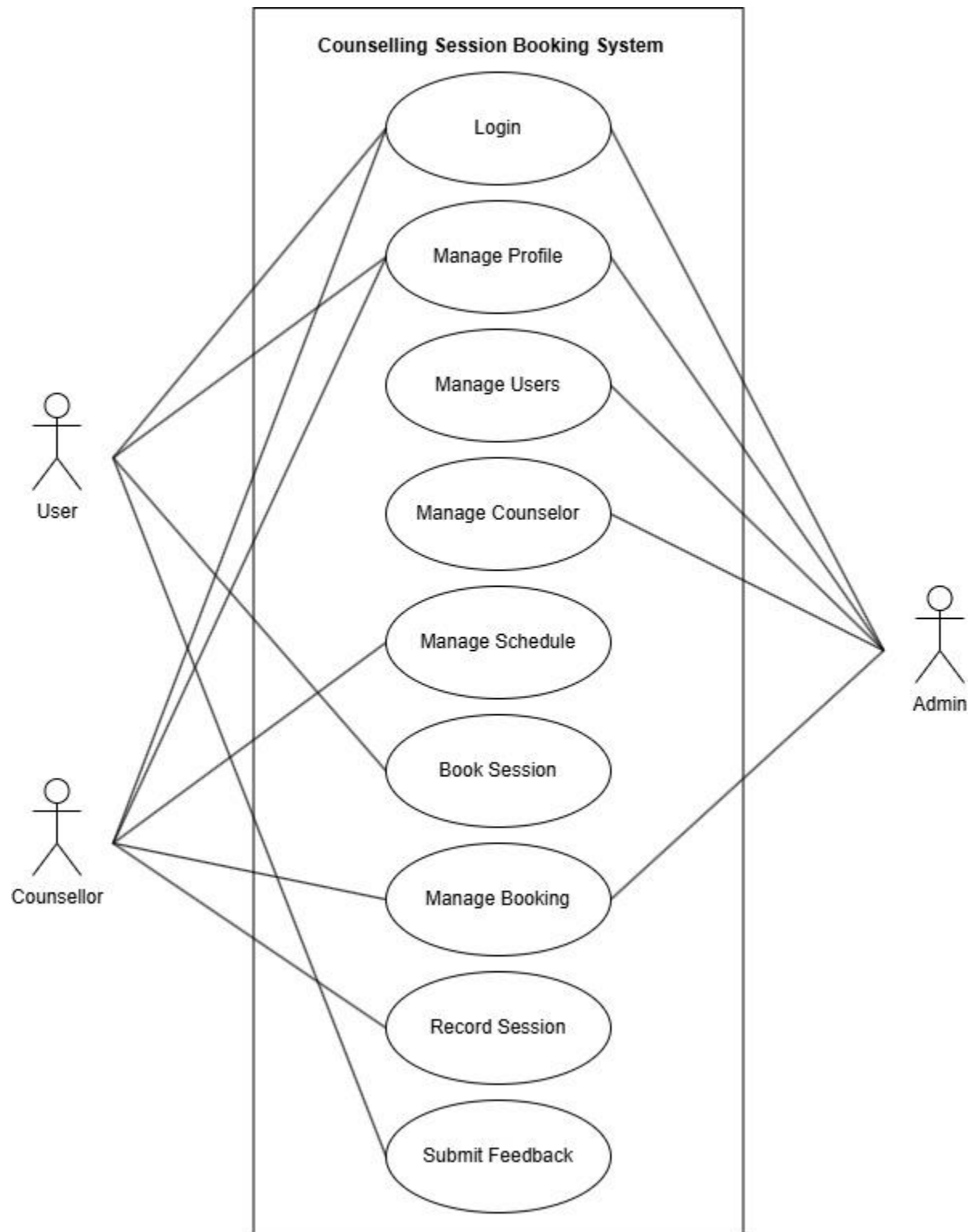
The project modules are distributed according to the system workflow. Each member is responsible for one specific module with CRUD functionality.

Group Number: Group 16

Member Name	Assigned Module	Module Functions (CRUD)
WAI PEI YUEN (Group Leader)	User Management, Schedule Management	User Management Create: register account View: login account Update: update profile Delete: delete account Schedule Management Create: create schedule View: view schedule Update: update schedule Delete: delete schedule
TENGKU AQIL BIN TENGKU AMDAN	Counsellor Management	Counsellor Management Create: add counsellor View: view counsellor list Update: update counsellor list Delete: remove counsellor
AMERRUL RAFIQUE BIN MOHAMED SULAIMAN	Booking Management	Booking Management Create: book session View: view bookings Update: update booking (reschedule) Delete: cancel booking
MUHAMMAD ADAM DANISH BIN MOHAMAD FAIZ	Session Record Management	Session Record Management Create: add session notes and feedback View: view session history Update: edit notes Delete: remove records

3.0 Use Case and Description

3.1 Use Case



3.2 Use Case Description

3.2.1 Login

Use Case Name: Login	ID: 1	Importance Level: High
Primary Actor: User, Counsellor, Admin	Use Case Type: Essential	
Stakeholders and Interests: User – Wants to access the system features Counsellor – Wants to manage schedule and sessions Admin – Wants to manage system operations		
Brief Description: This use case describes how users log in to the system using valid credentials.		
Trigger: User selects Login on the system homepage Type: External		
Relationships: Association: User, Counsellor, Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The User enters username and password. 2. The System validates the credentials. 3. The System grants access to the dashboard.		
SubFlows:		
Alternate/Exceptional Flows: 2a: Invalid credentials		

The system displays an error message and prompt re-entry.

3.2.2 Manage Profile

Use Case Name: Manage Profile	ID: 2	Importance Level: Medium
Primary Actor: User, Counsellor, Admin	Use Case Type: Essential	
Stakeholders and Interests: All Users – Wants to update their personal information easily		
Brief Description: This use case describes how users view and update their profile details.		
Trigger: User selects “Manage Profile” Type: External		
Relationships: Association: User, Counsellor, Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The User selects the profile section. 2. The System displays current information. 3. The User updates details. 4. The User submits changes. 5. The System validates and saves data.		
SubFlows:		
Alternate/Exceptional Flows: 5a: Invalid input The system displays an error message.		

3.2.3 Manage Users

Use Case Name: Manage Users	ID: 3	Importance Level: High
Primary Actor: Admin	Use Case Type: Essential	
Stakeholders and Interests: Admin – Wants to manage user accounts effectively		
Brief Description: This use case describes how the Admin manages user accounts.		
Trigger: Admin selects “Manage Users” Type: External		
Relationships: Association: Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Admin views the user list. 2. The Admin adds, updates or deletes users. 3. The System saves the changes.		
SubFlows:		
Alternate/Exceptional Flows:		

3.2.4 Manage Counsellor

Use Case Name: Manage Counsellor	ID: 4	Importance Level: High
Primary Actor: Admin	Use Case Type: Essential	
Stakeholders and Interests: Admin – Wants to manage counsellor information		
Brief Description: This use case describes how the Admin manages counsellor accounts.		
Trigger: Admin selects “Manage Counsellor” Type: External		
Relationships: Association: Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Admin views counsellor list. 2. The Admin adds, updates or deletes counsellor data. 3. The system saves the changes.		
SubFlows:		
Alternate/Exceptional Flows:		

3.2.5 Manage Schedule

Use Case Name: Manage Schedule	ID: 5	Importance Level: High
Primary Actor: Counsellor	Use Case Type: Essential	
Stakeholders and Interests: Counsellor – Wants to manage availability schedule		
Brief Description: This use case describes how the Counsellor sets and updates available time slots.		
Trigger: Counsellor selects “Manage Schedule” Type: External		
Relationships: Association: Counsellor Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Counsellor views current schedule. 2. The Cunsellor adds or updates time slots. 3. The system saves the schedule.		
SubFlows:		
Alternate/Exceptional Flows:		

3.2.6 Book Session

Use Case Name: Book Session	ID: 6	Importance Level: High
Primary Actor: User	Use Case Type: Essential	
Stakeholders and Interests: User – Wants to book a counselling session Counsellor – Wants organized schedule		
Brief Description: This use case describes how the User books a counselling session.		
Trigger: User selects available schedule Type: External		
Relationships: Association: User Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The User views available schedule. 2. The User selects a time slot. 3. The User confirms booking. 4. The System records the booking.		
SubFlows:		
Alternate/Exceptional Flows: 2a: Slot unavailable The system displays message.		

3.2.7 Manage Booking

Use Case Name: Manage Booking	ID: 7	Importance Level: Medium
Primary Actor: User, Counsellor, Admin	Use Case Type: Essential	
Stakeholders and Interests: Counsellor – Wants to manage session appointments Admin – Wants to oversee bookings		
Brief Description: This use case describes how bookings are managed.		
Trigger: User selects “Manage Booking” Type: External		
Relationships: Association: Counsellor, Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Actor views booking list. 2. The System updates booking status (approve/ cancel). 3. The System save changes.		
SubFlows:		
Alternate/Exceptional Flows:		

3.2.8 Record Session

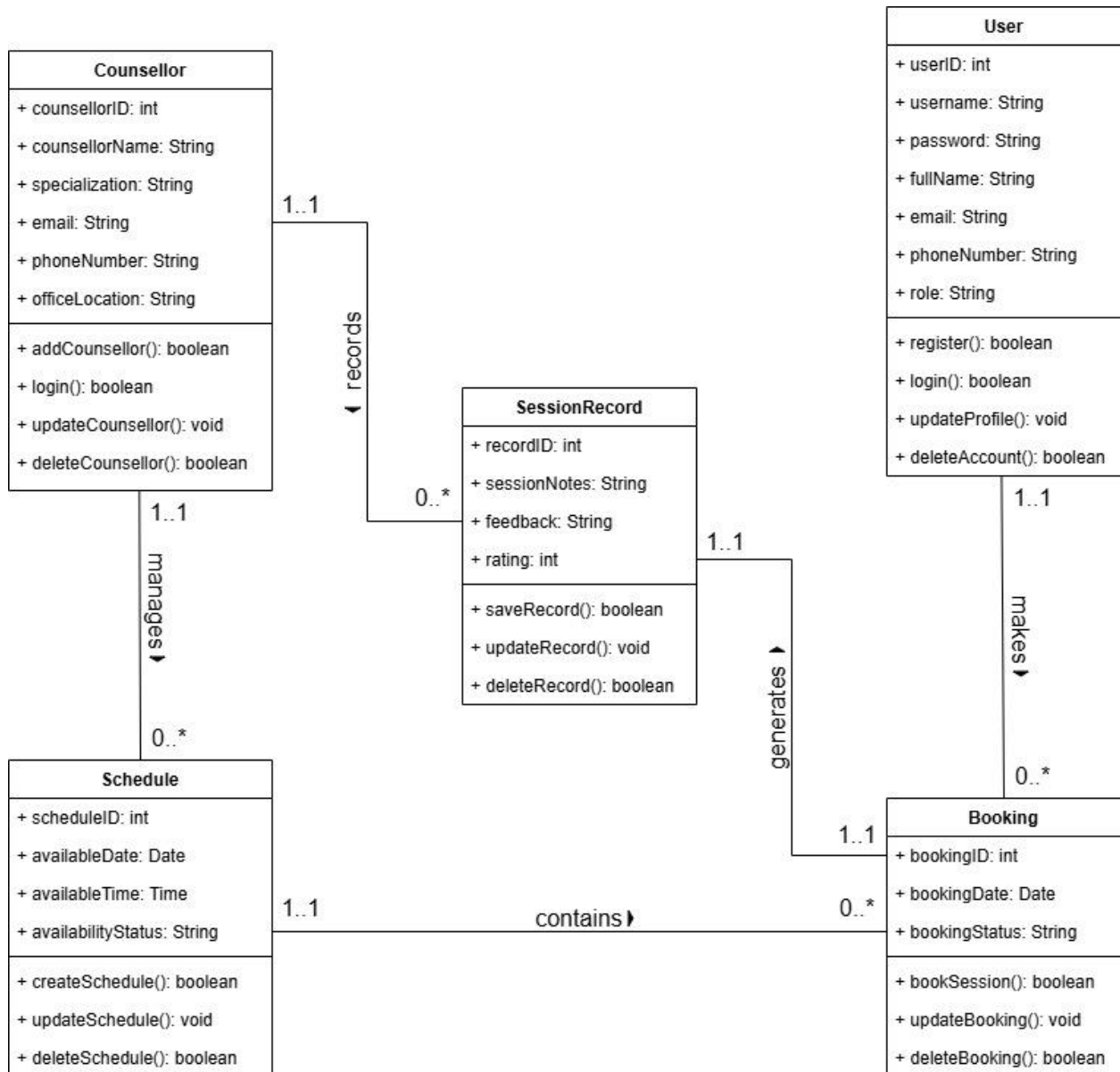
Use Case Name: Record Session	ID: 8	Importance Level: Medium
Primary Actor: Counsellor	Use Case Type: Essential	
Stakeholders and Interests: Counsellor – Wants to record session details		
Brief Description: This use case describes how the Counsellor records session notes.		
Trigger: Session completed Type: External		
Relationships: Association: Counsellor Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Counsellor selects completed session. 2. The Counsellor enters session notes. 3. The System saves the record.		
SubFlows:		
Alternate/Exceptional Flows:		

3.2.9 Submit Feedback

Use Case Name: Submit Feedback	ID: 9	Importance Level: Medium
Primary Actor: User	Use Case Type: Essential	
Stakeholders and Interests: User – Wants to give feedback Admin – Wants to improve service		
Brief Description: This use case describes how the User submits feedback.		
Trigger: User selects completed session Type: External		
Relationships: Association: User Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The User selects a session. 2. The User enters ratings and comments. 3. The System saves feedback.		
SubFlows:		
Alternate/Exceptional Flows:		

4.0 Class Diagram

4.1 Class Diagram



4.2 Database Planning (Model)

The database planning identifies all required data to be stored in the system together with the specific data types.

Table 1: users

Attribute	Data Type	Key	Description
user_id	INT	PK	Unique user ID
username	VARCHAR(50)	-	Username for login
password	VARCHAR(255)	-	User password
full_name	VARCHAR(100)	-	User full name
email	VARCHAR(100)	-	User email
phone_number	VARCHAR(20)	-	Contact number
role	VARCHAR(20)	-	User role (User/Admin)

Table 2: counsellors

Attribute	Data Type	Key	Description
counsellor_id	INT	PK	Unique counsellor ID
counsellor_name	VARCHAR(100)	-	Counsellor name
specialization	VARCHAR(100)	-	Counselling specialization
email	VARCHAR(100)	-	Counsellor email
phone_number	VARCHAR(20)	-	Contact number
office_location	VARCHAR(100)	-	Office location

Table 3: schedules

Attribute	Data Type	Key	Description
schedule_id	INT	PK	Unique schedule ID
counsellor_id	INT	FK	References counsellor
available_date	DATE	-	Available session date
available_time	TIME	-	Available session time
availability_status	VARCHAR(20)	-	Schedule status

Table 4: bookings

Attribute	Data Type	Key	Description
booking_id	INT	PK	Unique booking ID
user_id	INT	FK	References user
schedule_id	INT	FK	References schedule
booking_date	DATE	-	Date of booking
booking_status	VARCHAR(20)	-	Booking status

Table 5: session_records

Attribute	Data Type	Key	Description
record_id	INT	PK	Unique session record ID
booking_id	INT	FK	References booking
counsellor_id	INT	FK	References counsellor
session_notes	TEXT	-	Counselling session notes
feedback	TEXT	-	User feedback
rating	INT	-	User rating score
session_date	DATE	-	Date of counselling session

5.0 User Interface (UI) Design

5.1 User Interface (UI) Design

Tab

URL box

Login Page

Form: Login

Username

Text Box: Username

Password

Text Box: Password

Button: Login

Login Page

Tab

URL box

Navigation bar

Admin Dashboard

Button:
Show Users

Button:
Add User

Table: Users

Button:
Actions

Button:
Logout

Tab

URL box

Navigation bar

Add User Page

Form: Add User

Name:Text Box: Name

ID:Text Box: Matric Number

Phone Number:Text Box: Phone Number

Password:Text Box: Password

User Type:Dropdown: User Type

Button:
Save User

Button:
Add User

Button:
Logout

Tab

URL box

Navigation bar

Student Dashboard

Button:
Dashboard

Button:
Book Appt.

Button:
Feedback

Button:
Logout

Student Information

Name: -----
Matric Number: -----
Phone Number: -----

Booked Appointments

Table: Appointments

Buttons:
View Appt.

Tab

URL box

Navigation bar

Book Appointment

Button:
Dashboard

Button:
Book Appt.

Button:
Feedback

Button:
Logout

Form: Book Appointment

Date:

Date Picker: Date

Time:

Dropdown: Time

Counselor:

Dropdown: Counselor

Button:
Book Appt.

Tab

URL box

Navigation bar

Feedback

Button:
Dashboard

Button:
Book Appt.

Button:
Feedback

Button:
Logout

Form: Student Feedback

Table: Counseling Sessions

Text box:
Student Feedback

Button: Save Feedback

Tab

URL box

Navigation bar

Counselor Dashboard

Button:
Dashboard

Button:
Manage
Availability

Button:
Session
Report

Button:
Logout

Counselor Information

Name: -----
Counselor ID: -----
Phone Number: -----

Booked Appointments

Table: Appointments

Buttons:
View Appt.

Tab

URL box

Navigation bar

Manage Availability

Button:
Dashboard

Button:
Manage
Availability

Button:
Session
Report

Button:
Logout

Table: Availability

Date	Available	Times
	Checkbox: Available Day	Checkbox: Available Times

Tab

URL box

Navigation bar

Session Report

Button:
Dashboard

Button:
Manage
Availability

Button:
Session
Report

Button:
Logout

Form: Session Report

Table: Counseling Sessions

Text box:
Counselor
Feedback

Button: Save Feedback

5.2 Detailed Frontend Planning

HTML/JSP Planning

HTML and JSP will be used to create the structure and content of the web application pages. JSP will also allow dynamic data from the database to be displayed on the web pages.

- Login form
- Registration form
- Dashboard layout
- Booking forms
- Schedule tables
- Feedback forms

CSS Planning

CSS will be used to improve the appearance and user experience of the web application.

The system will use CSS for:

- Page layout and alignment
- Navigation bar design
- Button styling
- Form styling
- Table formatting
- Responsive design
- Colors and typography

JavaScript Planning

JavaScript will be used to handle client-side validation and interactive functions.

- Empty field validation
- Password validation
- Email format validation
- Confirmation messages
- Alert notifications
- Dynamic schedule selection

